A Positive-Mass History Interface for Nonlinear Dispersion

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Abstract

We package the nonlinear low/high dispersion split into the exact retained-history quantifiers required by adaptive stochastic domination.

For each positive-mass history h, let $q_L(h)$ be its low-dispersion conditional bound. Write $q_G(h)$ for the Gaussian high tail, $\ell(h)$ for the comparison loss, and $q_E(h)$ for the nonlinear chart complement.

Theorem 1. *If the retained state is conditionally sufficient and every positive-mass history satisfies the two normalized comparison premises, then*

$$q(h) \le q_L(h) + e^{\ell(h)} q_G(h) + q_E(h).$$

If the right side is uniformly at most q, the shared-uniform coupling gives Bernoulli domination and the previously certified connected-cluster bound.

Proof. Apply the low/high split and Stage 42 to each conditional fiber. Taking the supremum only after retaining the full state gives the row bound. The Stage 36 coupling applies to the complete positive-mass history table. □

The frozen centered-reference table lies strictly inside the continuous-channel budget and is cluster-subcritical.

Remark 1. *The physical table remains open because its density and complement premises have not been established uniformly over nonlinear retained histories.*